

Lab 1 - Product Description Outline

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Spoiler Alert

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1. Introduction

According to the FDA, foodborne illnesses cause over 48 million people to get sick, 128,000 people to be hospitalized, and the deaths of 3000 people every year. The FDA has also reported almost 300 recalls in the last year. You would think that with an issue that affects so many people, more would be on the lookout for updates regarding recalls, especially for consumable products. However, according to Dr. William Hallman, a speaker at the 2025 Food Safety Summit, only 13% of Americans check recall reports on federally mandated websites. Many people also have dietary restrictions, such as allergies and immunodeficiency, and need a quicker and easier way to get updates on food contaminated by products they can't have. Recalled products don't just affect the everyday person either. Some businesses lose millions of dollars in litigation due to unsafe products given to customers. The main issue here is that people aren't being notified in time, or sometimes at all, when they've purchased a recalled product. Most people don't get important notifications from the FDA, and the information that those who regularly check the FDA website gather usually spreads slowly. There isn't a product that gives consumers a personalized space to keep up with recent recalls. About 91% of Americans own smartphones, according to the Pew Research Center. We all check our phone notifications all the time, making this the quickest way to get updates on recalled products. Our mobile app, SpoilerAlert, will alert customers about recent recalls on products such as food, cosmetics, and pharmaceuticals. It will allow the user to choose if and how they want to receive notifications on recalled products they've bought or that have the potential to affect them by allowing them to personalize their accounts.

1.1 Problem

1.1.1 Problem Statement

1.1.2 Who is Affected

- Everybody
- Parents
- People with Allergens
- Immunocompromised People
- Companies

1.1.3 Problem Characteristics

- Awareness Gap
- Engagement Gap
- Customization Gap
- Urgency Gap

1.2 Solution

- Develop SpoilerAlert— a mobile and web application that provides personalized, real-time notifications of food recalls.

1.2.1 Solution Statement

- SpoilerAlert bridges the gap between recalls issuers (FDA, USDA, CSPP) and consumers by automating and customizing recall alerts.

1.3 Solution Characteristics

- Delivers real-time alerts through email or app notifications.
- Integrates multiple APIs (FDA, Barcode Lookup) for comprehensive recall detection.
- Supports barcode and receipt scanning for personalized recall matching.
- Stores user data securely for future recall cross-referencing.

2. Product Description

2.1 Key Product Features and Capabilities

- Spoiler Alert streamlines recall awareness with multiple methods of notification.
- Spoiler Alert has functionality to specialize the type of reports based on factors like:
Allergies and Location
- The user can scan receipts and barcodes for Spoiler Alert to track what recalls the user may be at risk of.
- Provide real-time recall notifications via email and mobile app alerts.
- Aggregates recall data from FDA, USDA, and CPSC databases to ensure complete and accurate information.
- Offers a personalized inventory summarizing all active recalls affecting the user's stored products or scanned items.
- Allows users to filter recalls by type or category (food, cosmetics, and medicine).
- Utilizes cloud-based data synchronization so that updates reach users across all their devices.
- Integrates a search and filter function for recalls by product name, brand, or UPC.

2.2 Major Components (Hardware/Software)

- App as the main user interface.
- Spoiler Alert Servers
 - Our servers grab data from the FDA API.
 - Our Servers try to pull information from the API to alert users about the recall.
 - Our servers will save previous user purchases to the servers to alert them about future recalls.

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- In the case that a UPC cannot be gathered from the recall, a Spoiler Alert employee will manually gather all relevant UPCs and link them.
- Include MFCD from the slides?

3. Identification of Case Study

3.1 Who is affected by recalls?: A case study

3.1.1 WHO

- A single mother of three, with two children who have severe allergies.

3.1.2 WHAT

- The problem: She struggles to monitor food recalls and misses critical alerts from the FDA.

3.1.3 HOW

- She'll use SpoilerAlert:

The single mother goes to the grocery store and buys some chicken to make for her and her children for dinner. She scans her receipt into the app after her purchase to log all of her groceries into her personalized inventory. The next day, she plans to use the chicken, but get's an alert from the app that it has been recalled due to salmonella. Luckily, she received this alert before her or her children consumed it.

3.2 FEATURES

- Personalized inventory in-app based on the users purchases that they can log manually or with scanning a barcode or receipt
- Notifications for recalls of any purchased products or logged allergies in whatever form the user decides

3.3 Future Capabilities

- Advertisements from sponsors for general users that stop when upgrading to premium
- The opportunity to opt into text messages for notifications

3.4 Other Potential Users

- Individuals who have dietary restrictions or have children with them
- Business owners and managers
- Immunocompromised individuals

4. Glossary

- Product Recall
- Recall Reviewer
- FDA
- CDC
- UPC
- CPSC
- API
- Allergen
- Immunocompromised
- Contamination

5. References

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